

TEST PROCEDURE FOR 180 kW ADLB CHARGERS [72115-0000-13-60_PLTP]

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1. Preface

1.1. Purpose

- This document describes the Testing Procedure for 180kW ADLB Fast DC Charger

1.2. Applicable Document

- GA (General Assembly): 72115-0000-18-02-R0_PLGA
- IA (Internal Assembly): 72115-0000-18-40-R0_PLIA
- Schematic: 72115-00-06-01_SCMT
- OCPP Testing Document
- Maintenance Mode Manual

1.3. Equipment Required

- Multi-Meter
- Electrical Tester
- Laptop
- Insulation Tester
- EV Test vehicle (or simulator)
- Personal protective equipment
- Ratchet and spanner

1.4. Main Components

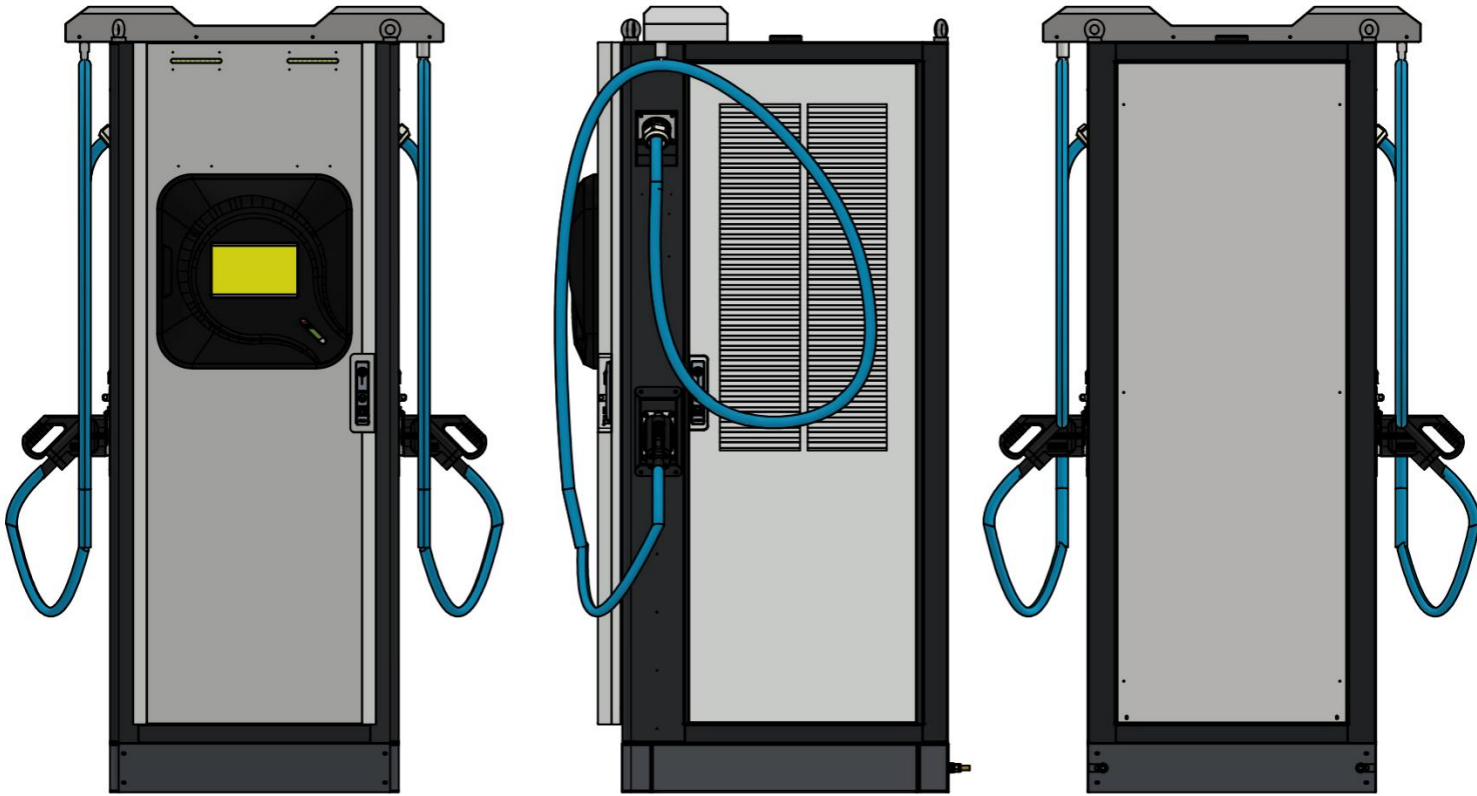
- MCCB (Moulded Case Circuit Breaker)
MCCB is mounted in the input AC mains line to make the Power ON/OFF to the system.
- ELR with CBCT
An Earth Leakage Relay (ELR) used with a Core Balance Current Transformer (CBCT) detects imbalance in phase currents and trips the circuit when leakage current exceeds the set limit, providing reliable earth-fault protection.
- SPD (Surge Protection Device)
SPD is used to protect system from over voltage and lightning spikes.
- Power Module
Each power module has capacity to deliver 30kW output. The power modules are used to convert 3 phase AC input to isolated DC output of up to 1000Vdc. The output current or power can be increase by paralleling of power module. In 180KW system, Six numbers of 30KW modules are used.
- AC Contactor
AC Contactor is an electrically controlled switch used to connect/disconnect mains supply to the power modules.
- DC Contactor
The main function of DC contactors is to connect/disconnect DC output of power modules from output Guns.

- **ACEM (AC Energy Meter)**
ACEM measures multiple AC Input characteristics. Its main purpose is to record the energy used by the charger over time and monitor other parameters like L-L Voltage, Current, power factor.
- **DCEM (DC Energy Meter)**
DCEM measured multiple DC output characteristics required for better billing accuracy. It monitors multiple parameters like DC voltage, current, power and mainly energy consumption.
- **IMD (Insulation Monitoring Device)**
IMD continuously checks the insulation resistance between positive/negative poles and ground of both CCS2 DC guns. Whenever the insulation resistance drops below the threshold, alarm is triggered and charging stops.
- **UPS (Power backup)**
In case of power failure. This keeps the controllers and HMI on to finish the billing process.
- **Control Circuit**
Control circuit is a combination of SECC2300, CPPP2300, SECCLE2300 and application board which are used to control the Charger outputs as well as continuously check the protections. It also communicates with electric vehicle as per the communication protocols for CCS2. It is also capable to communicate with HMI, RFID cards and communicate with CMS (Central management server over OCPP protocol) via Internet.

2. Visual check

2.1. Mechanical Assembly check

- Ensure the fabrication is dent-free and with proper paint as per figure 1



- Check stickers (standard and customer-specific stickers) are present as per AWI and customer specific requirements.
- Ensure labelling stickers for all components inside the panel
- Cross check all the mechanical assembly is completed as per the mechanical TC and connections are tightened.

2.2. Power wiring Check

- Check wire routing and connection of Input busbar, MCCB and AC Contactor as per Figure 2

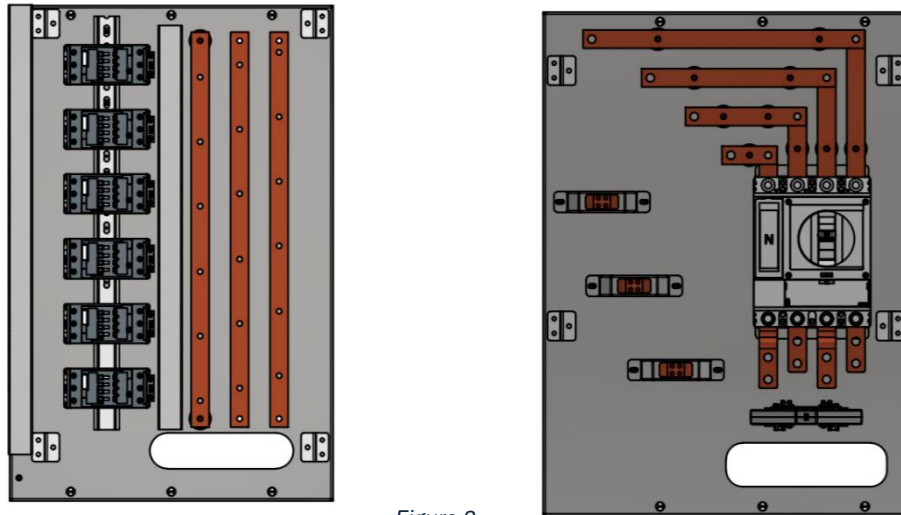


Figure 2

- Verify that the connections at the SPD, ELR, MCB, AC and DC contactors, and the input/output of power modules have proper termination lugs and are securely tightened.
- Ensure the output from each module is correctly routed to the respective DC contactor and the respective GUN
- The earth connection should be made to the busbar and distributed directly to all respective points. The panel body must also be connected to earth busbar.
- All hardware should be tightened with the specified torque as required. Ensure paint mark on hardware used for power wiring for AC and DC both.
- All cables should have proper stress relief

2.3. Control wiring Check

- Check all control wires, including those for logic, controllers, fan control, SMPS 1 and 2 (inputs/outputs), relays, contactor coils and IMD, display, and power supply, AC Terminal TB and DC terminal TB and AC Meter and DC Meter as per the Schematic.
- Verify the cable routing, lugs for control cables, and connectors as per harnessing and AWI (Assembly work Instructions)
- Check all controller connectors are fitted well. No loose connections allowed.

2.4. Communication wiring Check

- Check the communication wiring as per schematic
 - Ensure shielded wire used along with heat shrink sleeve (at points of cable stripping) for following connections:
 - UNITRONIC Bus CAN Cable

- Controller to Power module- CAN cable
- Ensure 120Ω resistor present at start (SECCLE) and end of circuit (PM2)
- 2 core industrial cable with tinned copper braiding (80%)
 - Controller to Insulation monitoring device
 - Controller to AC meter
 - Controller to DC meter Ch1 & Ch2
 - Controller to Gun A CP and Gun B CP
 - Ensure 120Ω resistor present in RS485 Bus connections at both ends of wire termination
- Check tightness and correct cable sequence for RJ45 Cable
 - Router to SECC
 - Router to SECC-LE
 - Router To AB

3. Megger/Insulation Resistance and HV Test

3.1. Megger/Insulation Resistance Test

- Instrument Required: Insulation Tester with 500V capacity.
- Test procedure
 - Disconnect PE from
 - Controller
 - Router
 - SMPS1,2
 - IMD
 - SPD
 - Power Module
 - GNDF
 - Charger Main Earthing
 - Disconnect Input Power Supply
 - Short AC Input Terminal Busbar (R,Y,B,N) and DC output (+,-) of both guns
 - Apply Insulation test voltage between the shorted AC inputs and the Earth Terminal.
 - Apply Insulation test voltage between the shorted DC Output (+,-) terminals and the Earth Terminal.
 - **Passing criteria: Resistance should be $\geq 10M\Omega$**

3.2. HV Test

- Instrument Required: HV Tester with 1.5kV capacity
- Test Procedure
 - Disconnect PE and Input power supply from:
 - Controller
 - Router
 - SMPS1,2,3

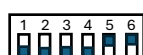
- IMD
- SPD
- Power Module
- GNDF
- Charger Main Earthing
- Short AC Input Terminal Busbar (R,Y,B,N)
- Apply HV between shorted AC inputs and the Earth terminal
- Perform Tests at the 1440 V According to ISI 17017
- **Passing Criteria: Leakage Current should $\leq 5\text{mA}$**

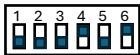
4. Electrical Testing

4.1. Pre-Check (Continuity and Earthing Tests)

- Check all fuse ratings.
- Remove all Short links in 8 channel Relay Base.
- Use a multi-meter to check continuity for the following:
 - Check short links present between individual terminal Blocks for XL, XN, 24V1,0V1, 24V2, 0V2
 - Check all earthing connections continuity
 - Check SMPS1 and SMPS2 output supply wiring continuity
 - Check AC Meter V1, V2, V3, N to MCCB L1, L2, L3, N Continuity
 - Ensure there shall not be any short circuit 24V1 & 24V2.
- Open all fuses and remove input supply for the following components (AB, HMI and SECC J1 and CPPP J6)
- Connect 3 phase supply to MCCB and check voltage with multimeter then turn on MCCB
- Check AC input voltage and DC output voltage for SMPS 1 & 2 and close the respective fuses.
- Check respective voltages at:
 - DC jack cable for AB = 5V
 - DC jack cable for HMI = 24V
 - SECC_J1 = 24V
 - CPPP_J8 = 24V
 - SECC-LE_J1 = 24V
 - DC meter = 24V
- If voltages are correct, connect all supply connections and check whether components turn on
- Check touch screen functionality

4.2. Configuration

- **Set Power Module address**
 - Power Module 1: Binary Number 0 default address set
 - Power Module 2: Need to set address binary Number 1 
 - Power Module 3: Need to set address binary Number 2 
 - Power Module 4: Need to set address binary Number 3 

- Power Module 5: Need to set address binary Number 4 
- Power Module 6: Need to set address binary Number 5 

- **ELR Configuration**

- **Set leakage current: 0.1 I Δn (A)**
- **Set Time: 1 Δt (S)**

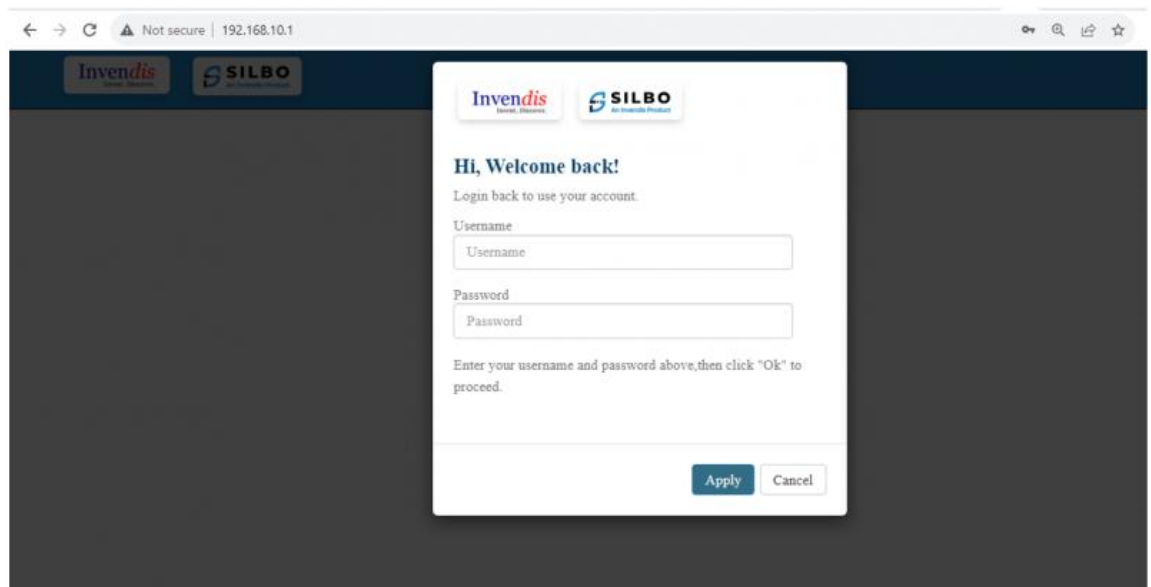


- **Router Configuration**

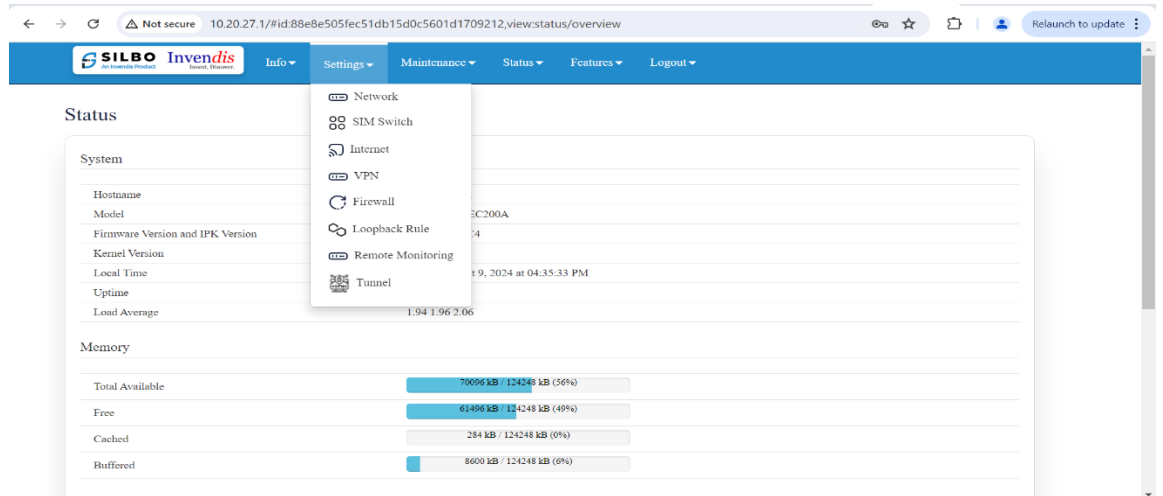
Model: RB44

Make: Silbo

- Please check SSID mentioned at the back of the router for the network name
 - Connect laptop to the router either via LAN cable or via wifi by connecting to the SSID mentioned
 - The network name will most likely start with “AP_” .
 - Enter password: 12345678
 - Once connected to the network, enter default URL 192.168.1.1
 - Enter Username: **admin** and password: **admin** on the following page

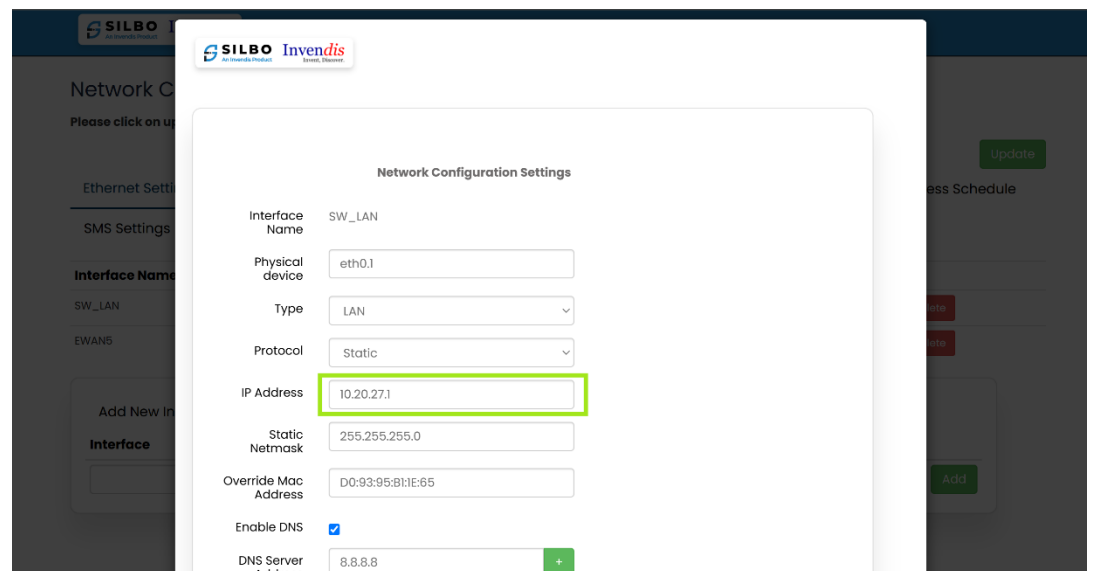


Go to Setting > Network

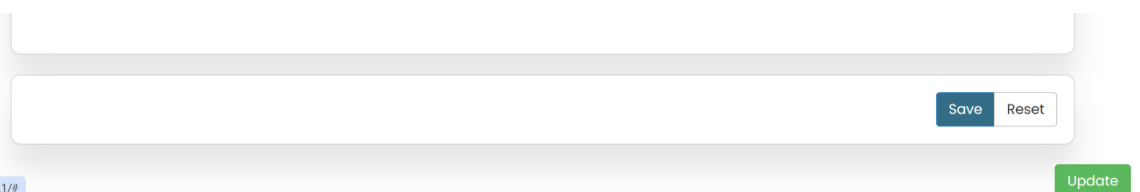


Network > Ethernet setting

Enter IP "10.20.27.1"



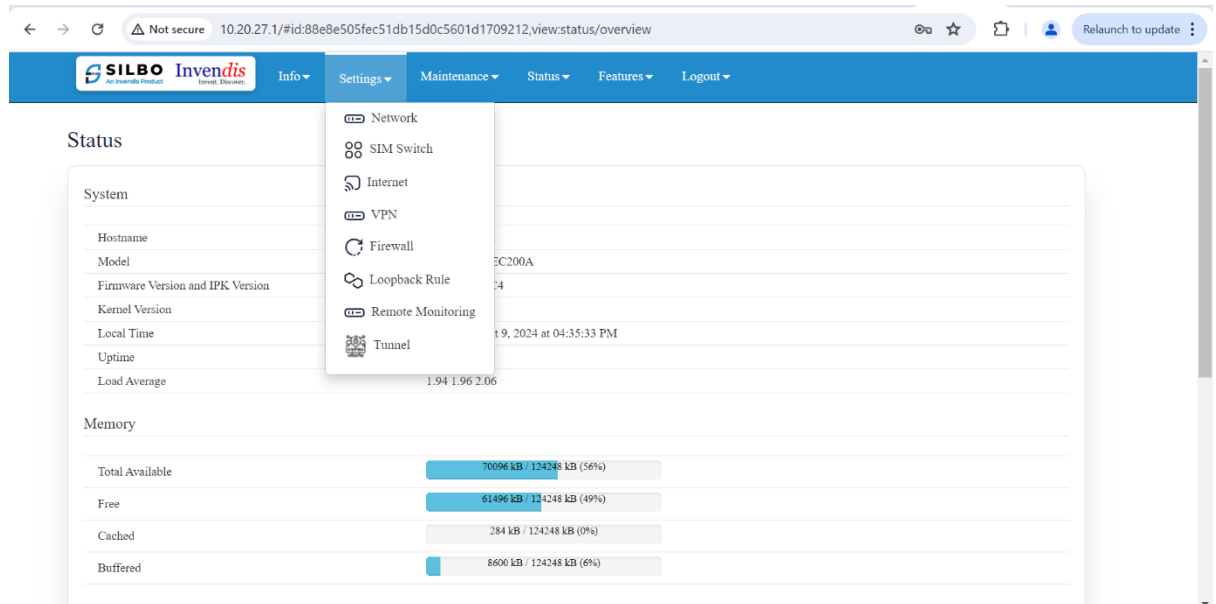
Press save and then update



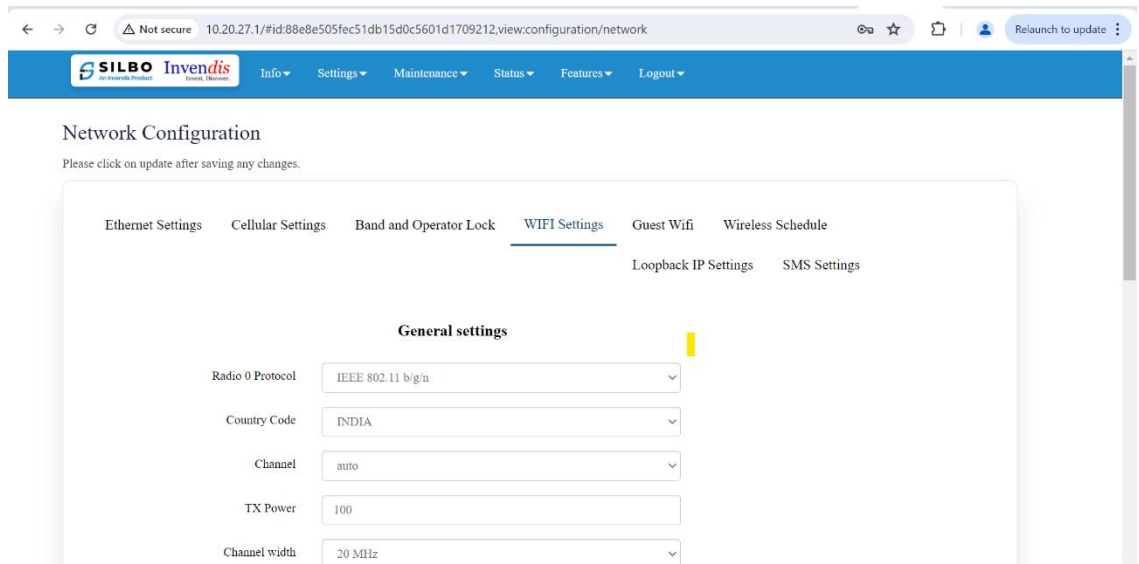
10.20.27.1/#

Update

- Check router configuration
 - If router was successfully updated in the previous step, open new browser/window and enter **URL: 10.20.27.1**
 - Enter Username: **admin** and password: **admin**
 - Below screen should appear:
 - Go to Settings→ Network



- Following Screen will appear the go to WIFI Settings



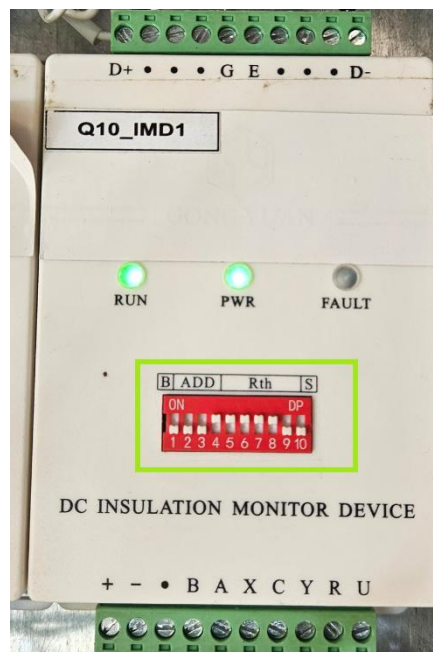
- Just scroll down and enter following highlighted field related to WIFI Network to be connected [WIFI Name & password]
- Press Save and Update. The router should now be connected to the internet.

The screenshot shows the 'view:configuration/network' page of the SILBO InvenDis web interface. The browser address bar shows '10.20.27.1/#id:88e8e505fec51db15d0c5601d1709212'. The interface has a blue header with navigation tabs: Info, Settings, Maintenance, Status, Features, and Logout. The main content area lists various network settings:

- Radio Mode: Access Point and Client
- Radio SSID: SILBO_RB44_31D50230081
- Radio Authentication: WPA Personal (PSK)
- Radio Encryption: TKIP
- Radio Passphrase: [Redacted]
- Radio DHCP Server IP: 192.168.100.1
- Radio DHCP Start Address: 100
- Radio DHCP Limit: 50
- Client SSID: Phone_1_5479 (highlighted with a green box)
- Client Encryption: WPA Personal (PSK)
- Client Passphrase: [Redacted] (highlighted with a green box)

• IMD Configuration

- Set same address on both IMD as mentioned below
 - There are 10 Dip switch
 - Enable 4 to 8 Dip switch



- **AC Meter configuration**

- Set Primary CT Current Rating 200A according to below mention procedure

- **AC meter Key Functionality**

There are 6 dedicated keys with symbols marked as , , , , , . use these 6 keys to enter into configuration menu / change setting.

Note : The settings should be done by a professional, after going through this users manual and after having understood the application situation.

For the configuration setting mode :

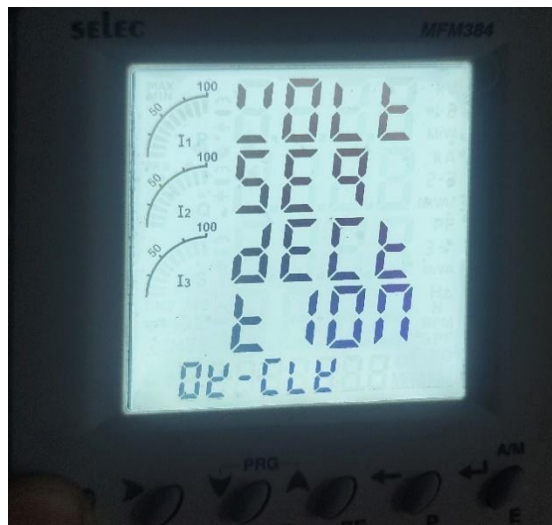
- Use  +  keys for 3 sec. to enter or exit from the configuration menu.
- Use  or  keys to move cursor left or right by one digit each time.
- Use  or  keys for increasing or decreasing parameters value.
- Use  key to go back to previous page.
- Use  key to save the setting and move on to next Page.

- **Parameter setting**

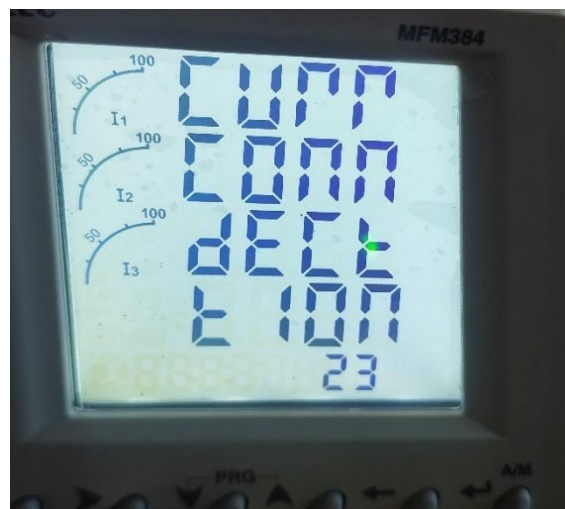
Config page.	Function	Range or Selection	Factory Setting
	Password	0000 to 9998	1000
1	Change Password	No / Yes	No
1.1	New Password	0000 to 9998	1000
2	Network Selection	3P4W, 3P3W, 1P2W-P1, 1P2W-P2 and 1P2W-P3	3P4W
3	CT Secondary	1A or 5A	5
4	CT Primary	1A, 5A to 10,000A	5
5	PT Secondary	100V to 500V	350
6	PT primary	100V to 500kV	350
7	Slave Id	1 to 255	1
8	Baud Rate	300, 600, 1200, 2400, 4800, 9600 and 19200 (bps)	9600
9	Parity	None, Odd, Even	None
10	Stop Bit	1 or 2	1
11	Back Light	0 to 7200 sec.	0000

Config page.	Function	Range or Selection	Factory Setting
12	Demand interval method	Sliding / Fixed	Sliding
13	Demand interval duration	1 to 30	15
14	Demand interval length	1 to 30 min	1
15	Max Page Auto	1 to 21	21
16	Change Page Sequence	No / Yes	No
16.01	Page sequence 1	1 to 21	1
16.02	Page sequence 2	1 to 21	2
16.03	Page sequence 3	1 to 21	3
16.04	Page sequence 4	1 to 21	4
16.05	Page sequence 5	1 to 21	5
16.06	Page sequence 6	1 to 21	6
16.07	Page sequence 7	1 to 21	7
16.08	Page sequence 8	1 to 21	8
16.09	Page sequence 9	1 to 21	9
16.10	Page sequence 10	1 to 21	10
16.11	Page sequence 11	1 to 21	11
16.12	Page sequence 12	1 to 21	12
16.13	Page sequence 13	1 to 21	13
16.14	Page sequence 14	1 to 21	14
16.15	Page sequence 15	1 to 21	15
16.16	Page sequence 16	1 to 21	16
16.17	Page sequence 17	1 to 21	17
16.18	Page sequence 18	1 to 21	18
16.19	Page sequence 19	1 to 21	19
16.20	Page sequence 20	1 to 21	20
16.21	Page sequence 21	1 to 21	21
17	Pulse weight (kWh)	0.01 to 99.99	0.01
18	Pulse duration (Sec)	0.1 to 2.0	0.1
19	Factory default	No / Yes	No
20	Reset energy and MAX demand	No / Yes	No
20.1	Password	0001 to 9999	1001
20.01	Reset active energy	No / Yes	No
20.02	Reset reactive energy	No / Yes	No
20.03	Reset apparent energy	No / Yes	No
20.04	Reset Current MAX DMD	No / Yes	No
20.05	Reset active power MAX DMD	No / Yes	No
20.06	Reset active power MIN DMD	No / Yes	No
20.07	Reset Reactive power MAX DMD	No / Yes	No
20.08	Reset Reactive power MIN DMD	No / Yes	No
20.09	Reset apparent power MAX DMD	No / Yes	No
21	Reset Run Hour	No / Yes	No
22	Reset Auxiliary Interrupt	No / Yes	No

- Hold 'V' button for 3 seconds and check phase sequence. It should read 'OK – CLK'



- Hold 'I' button for 3 seconds and check current connection indication page. It should read '23'. This means the current direction for L2 and L3 is reversed which is **REQUIRED**.



- If holding 'I' button reads '0', it means all 3 line current directions are OK. So we would have to interchange S1 & S2 connections (CT) for L2 and L3 such that their current direction is reversed. After interchanging the respective connections, reboot the AC Meter and hold 'I' button for 3 seconds to check current connection indication page. It should now read '23'.
- For instance, if one presses 'I' button for 3 seconds, and it reads '13'. This means current directions for L1 and L3 are reversed. We would have to interchange S1 & S2 connections for L1 and L2 such that they get reversed as we require the connections

reversed only for L2 and L3. After rebooting AC Meter and checking current connection indication page, it should now read '23'.

- Open SECC and check IOMapper.
- During Idle conditions:
 - Avg PF should be positive.
 - Active power for L1 should be positive
 - Active power for L2 should be negative
 - Active power for L3 should be negative

```

{ 1 item
  "modbus": { 1 item
    "selec": { 33 items
      "active_L1": 0.06421352922916412
      "active_L2": -0.014053067192435265
      "active_L3": -0.022828446701169017
      "active_export_total": 0
      "active_import_total": 1
      "active_total": 0.027332015335559845
      "active_total_KVA": 1.711665153503418
      "apparent_L1": -0.5436959266662598
      "apparent_L2": -0.5719202756881714
      "apparent_L3": -0.5916571021080017
      "apparent_total": -1.707273244857788
      "average_pf": 0.01600000075995922
      "current_L1": 2.2839999198913574
      "current_L2": 2.3959999084472656
      "current_L3": 2.4600000381469727
    }
  }
}

```

- During charging:
 - avg PF should be positive and close to 0.9
 - Active power for L1 should be positive
 - Active power for L2 should be positive
 - Active power for L3 should be positive
 - Only active_import_total energy should be increasing.
 - Export energy should remain 0

```

{ 1 item
  "modbus": { 1 item
    "selec": { 33 items
      "active_L1": 1.5758007764816284
      "active_L2": 1.2555296421051023
      "active_L3": 1.2865419387817385
      "active_export_total": 0
      "active_import_total": 1
      "active_total": 4.11787223815918
      "active_total_KVA": 4.552897930145264
      "apparent_L1": -0.4501730501651764
      "apparent_L2": -0.6962841153144836
      "apparent_L3": -0.7282958030700684
      "apparent_total": -1.8747529983520508
      "average_pf": 0.910099983215332
      "current_L1": 6.868000030517578
      "current_L2": 6.0320000648498535
      "current_L3": 6.171999931335449
    }
  }
}

```


- **DC Meter Configuration**

- Modbus Address by default  Set Add-002
- Parity by default  Set PAR - None
- Shunt Current rating by default  Set SH1- 500A , SH 2- 500A


- **DC Meter key functionality**



Indicates paging up when switching display pages, left shifting when inputting number, return when viewing parameters.



Indicates paging down when switching display pages, increasing when setting parameters.

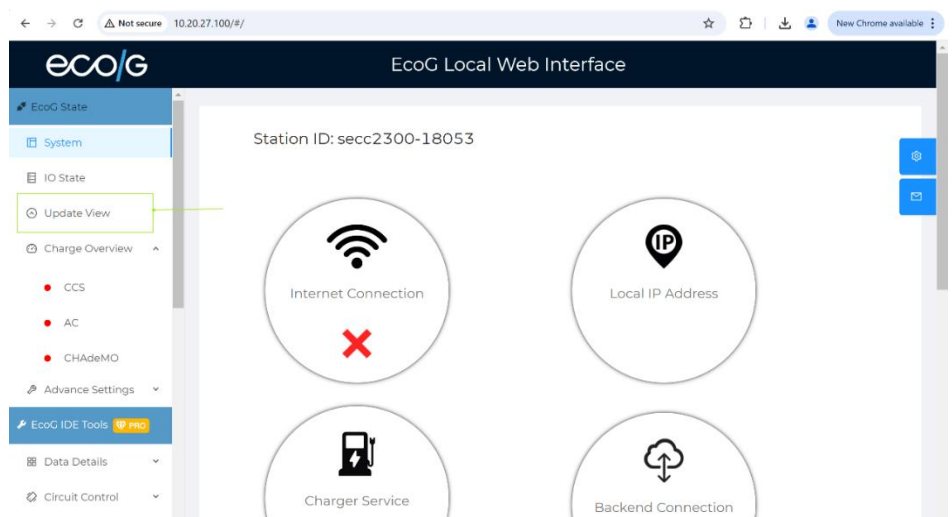


Indicates confirming.

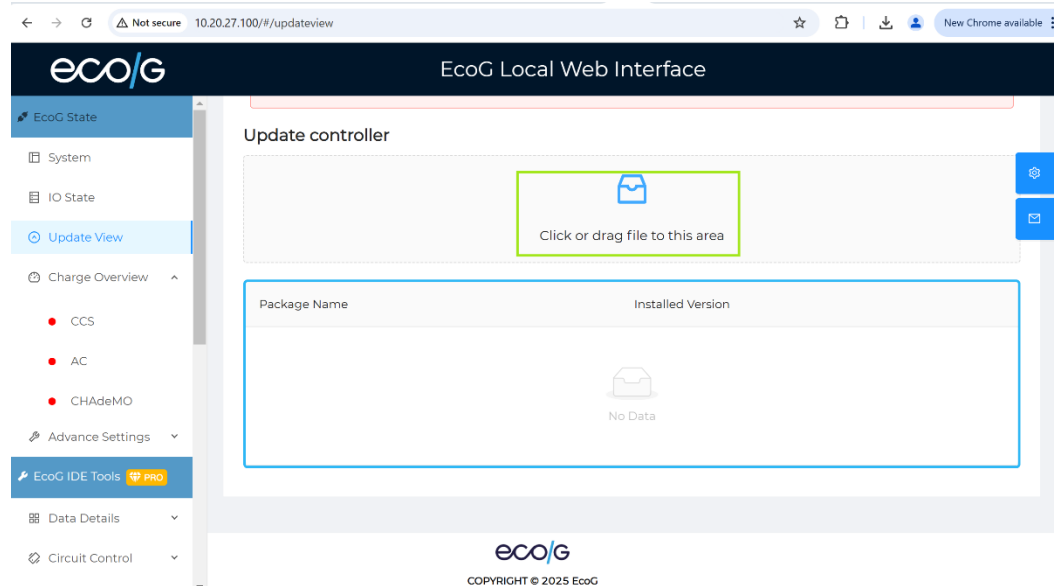
- Step1. Press the  button to open the Password Input Interface (PSA)
- Step2. Enter password: 1101 to access the Configuration Page.
- Step3. Press the  button to switch between different setting items.
- Step4. Press the  button to edit setting parameter item flashes Enter above mention values Modbus Address, Parity, Shunt Rating.

- **Update and Configure Controller and AB**

- **Update Controller**
- Open LWI on laptop (SECC) URL "10.20.27.100" Enter on browser
- Update latest Release
 - Step1. Click on update view

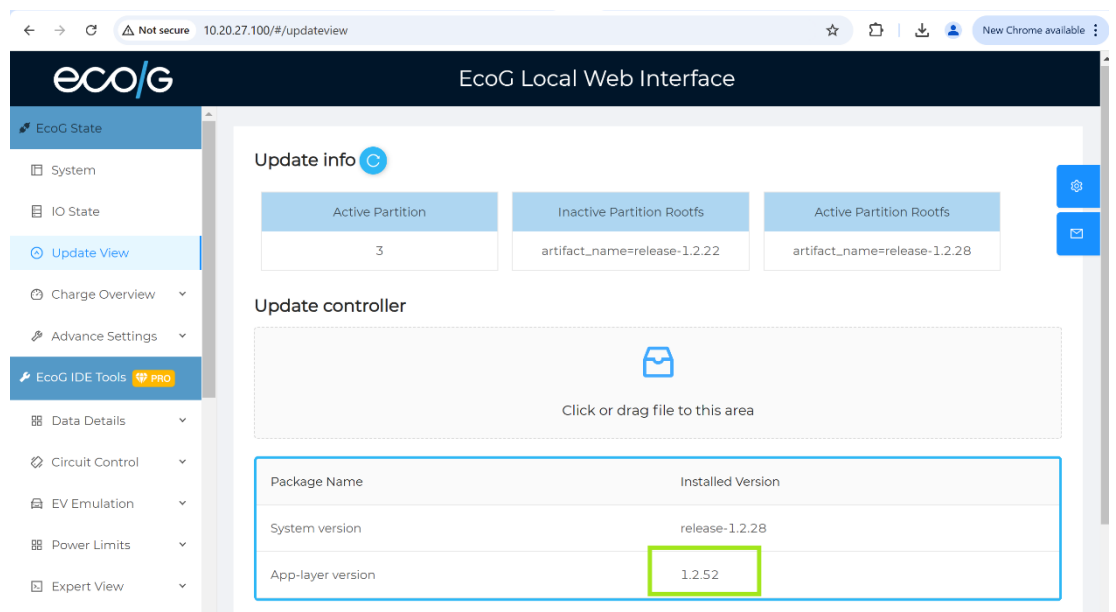


- Step2. Click on Highlight area select SECC Ecog update file and start Update



- Step3. Let update complete 100% and Click on Reboot
- Step 4. Check firmware of release

- Open LWI on Laptop (SECC-LE) URL "10.20.27.101" Enter on browser



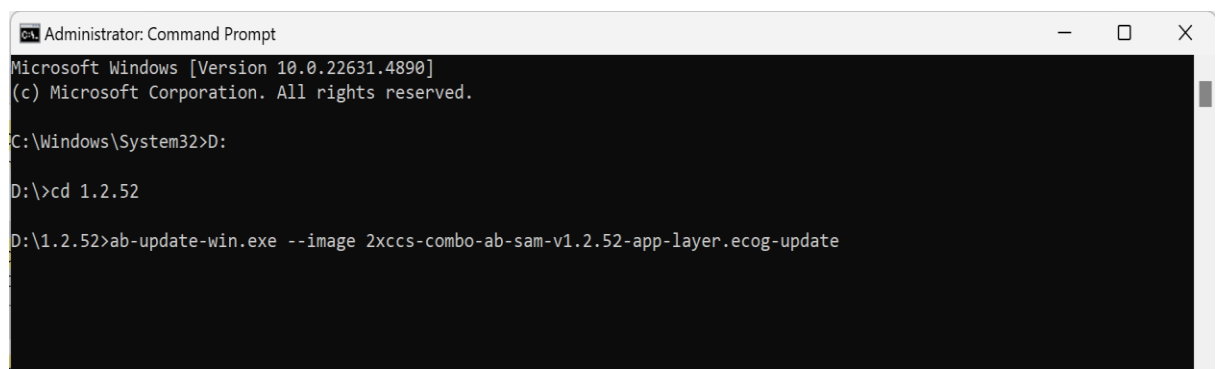
- Update SECC-LE following above mention steps
- For SECC-LE update select SECC-LE Ecog update file
- Let update complete 100% and Click on Reboot

- Configure SECC and SECC-LE Controller
 - Open SECC LWI URL "10.20.27.100"

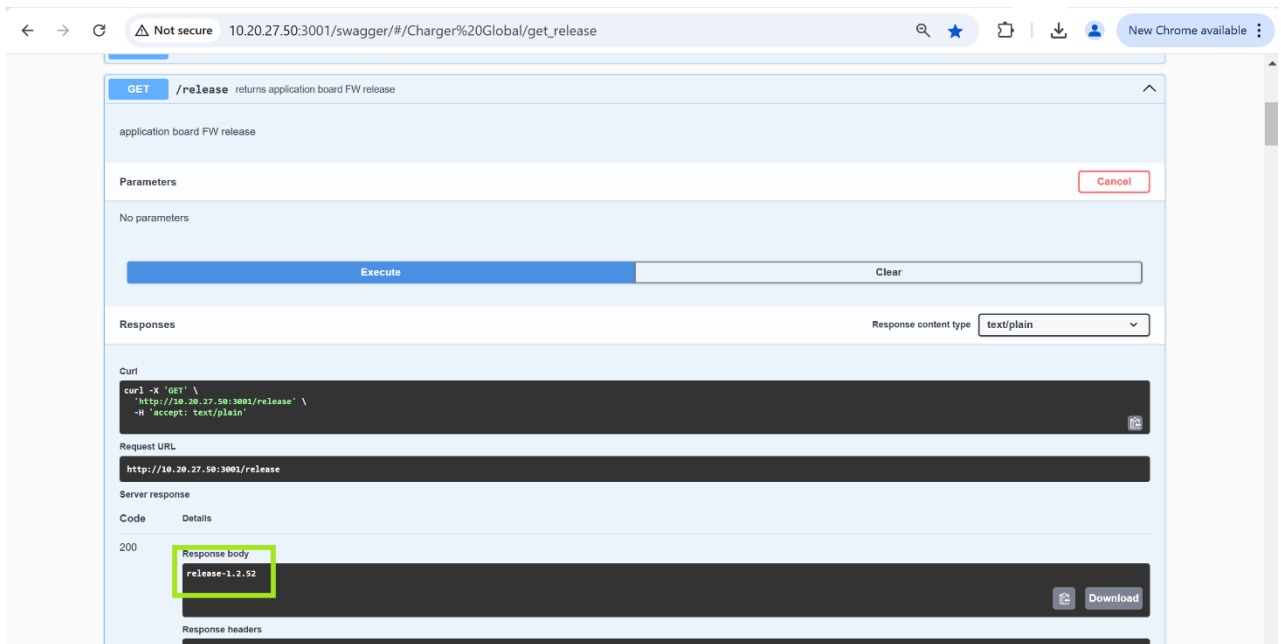
- Open SECC-LE LWI URL "10.20.27.101"
- Follow same steps to configure SECC & SECC-LE
- Go to advance > Configuration
- Change num_of_module to **6**
- Change maxkW to **=180 and maxA to 500**
- Change dlbmode to **quintupleCombo**
- Click On Load to device



- **Update AB**
- Open Swagger URL "10.20.27.50:3001" Enter on browser
 - Step1. Right click on CMD select Run as administrator
 - Step2. Type **D:** and Press Enter > Type **cd** > Press space
 - Step3. Press tab to select AB release folder > Press Enter
 - Step4. Press tab to select **(ab-update-win.exe)** > Type --image press tab Select AB ecog-update file and Enter

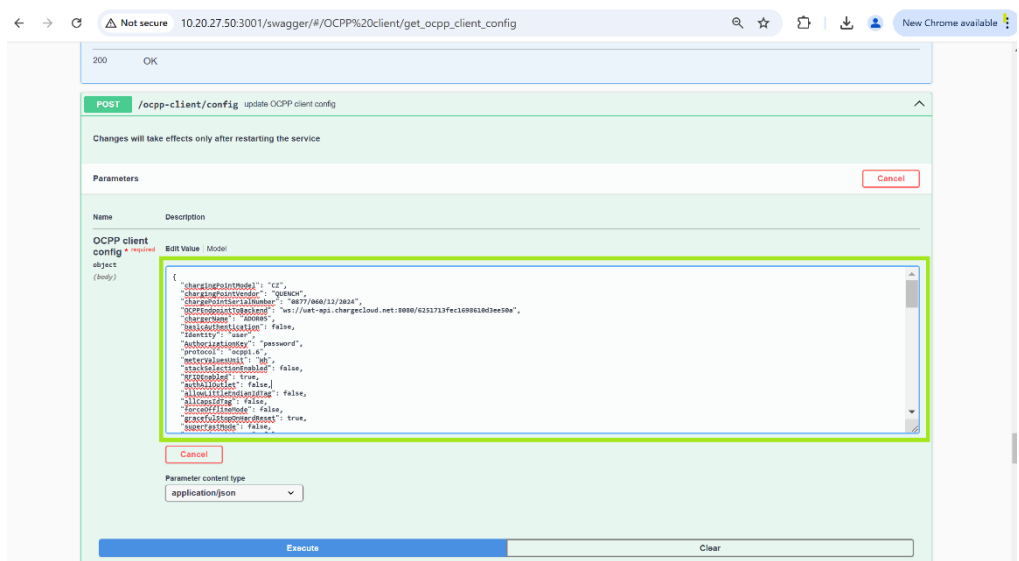


- Step3. Wait for “installation successful “AB Reboot.
- Step4. Check firmware of updated Release



○ Configure AB

- Open Swagger URL “10.20.20.27.50:3001” on Browser
 - Go to Post `/ocpp-client/Config` > click on try out
- Configure config Key as per Customer requirements
 - Download response body code and Edit below mention key
 - Change `"chargingPointModel"`: **"Customer Name"**,
 - Enter `"chargePointSerialNumber"`: **"Serial No."**,
 - Change `"powerSaveIdleMode"`: **true/ false** , as per requirement
 - Copy whole Changed code
 - Go to Post `/ocpp-client/Config` > try it out
 - Paste copied Code in Highlighted area > click on execute



Example OCPP Configuration keys for reference:

```
{
  "chargingPointModel": "CZ",
  "chargingPointVendor": "QUENCH",
  "chargePointSerialNumber": "0877/060/12/2024",
  "OCPPEndpointToBackend": "ws://uat-api.chargecloud.net:8080/6251713fec1698610d3ee50a",
  "chargerName": "ADOR05",
  "basicAuthentication": false,
  "Identity": "user",
  "AuthorizationKey": "password",
  "protocol": "ocpp1.6",
  "meterValuesUnit": "Wh",
  "stackSelectionEnabled": false,
  "RFIDEnabled": true,
  "authAllOutlet": false,
  "allowLittleEndianIdTag": false,
  "allCapsIdTag": false,
  "forceOfflineMode": false,
  "gracefulStopOnHardReset": true,
  "superFastMode": false,
  "PreparingStatusDC": false,
  "customMeasurands": false,
  "sendEventAsDataTransfer": false,
  "maxPowerLimitInkW": 80,
  "maxCurrentLimitInAmps": 200,
  "powerSaveInIdleMode": true,
  "autoChargeMode": true,
  "underVoltageThreshold": 353,
  "overVoltageThreshold": 477,
  "emulatedMetering": true,
  "storeSessionAndEvent": true,
  "singleTimeStampOnMeterValueMeasurand": false,
  "OCPPBackendConnectionTimeout": -1,
  "devSpecificFeatures": false,
  "standard": {
    "StopTxnSampledData": "Energy.Active.Import.Register, SoC",
    "ConnectionTimeOut": 60,

```

```

"ClockAlignedDataInterval": 0,
"MeterValuesAlignedData": "Energy.Active.Import.Interval",
"MeterValuesAlignedDataMaxLength": 5,
"AllowOfflineTxForUnknownId": false,
"AuthorizationCacheEnabled": false,
"AuthorizeRemoteTxRequests": true,
"GetConfigurationMaxKeys": 100,
"HeartbeatInterval": 300,
"LocalAuthListEnabled": true,
"LocalAuthorizeOffline": true,
"SupportedFeatureProfiles": "Core, FirmwareManagement, LocalAuthListManagement, Reservation, RemoteTrigger, SmartCharging",
"LocalAuthListMaxLength": 10000,
"SendLocalListMaxLength": 2500,
"LocalPreAuthorize": true,
"MeterValueSampleInterval": 60,
"MinimumStatusDuration": 1,
"WebSocketPingInterval": 5,
"TransactionMessageAttempts": 5,
"TransactionMessageRetryInterval": 20,
"MeterValuesSampledData": "Energy.Active.Import.Register, Power.Active.Import, SoC, Current.Import, Voltage, Current.Offered, Power.Active.Import, Temperature, Power.Offered, Energy.Active.Import.Interval",
"ChargeProfileMaxStackLevel": 20,
"ChargingScheduleAllowedChargingRateUnit": "Power",
"ChargingScheduleMaxPeriods": 10,
"ConnectorSwitch3to1PhaseSupported": false,
"MaxChargingProfilesInstalled": 10,
"ReserveConnectorZeroSupported": false,
"StopTransactionOnInvalidId": false,
"NumberOfConnectors": 2,
"ConnectorPhaseRotation": "NotApplicable",
"ResetRetries": 4,
"StopTransactionOnEVSideDisconnect": true,
"StopTxnAlignedData": "Energy.Active.Import.Register, SoC",
"UnlockConnectorOnEVSideDisconnect": true,
"BasicAuthPassword": "password",
"HeartBeatInterval": 30
},

```

```

"state": {
"meterValue": "curr_ses_Wh",
"firmwareVersion": "version",
"identifier": "user",
"energyActiveImportRegister": "curr_ses_Wh",
"energyActiveImportRegisterAC": "aux/ac_meter"
},
"dataTransfer": {
"VendorsId": [
"MSIL"
],
"MessagesId": [
"AutoStop"
]
},
"vendorsId": []
}

```

4.3. Communication Testing

- Check Module communication
- Open SECC LWI
 - Go to IO Mapper > click on data
- Check Selec AC meter Communication
 - Check "modbus.selec.online":true
 - Check AC Parameter
 - 3 phase Voltage and Current
 - Ensure actual value and display value are same. (V, I, PF etc)
 - Check all configure parameters shows in IO mapper

"modbus.selec.active_L1":0
"modbus.selec.active_L2":0
"modbus.selec.active_L3":0
"modbus.selec.active_export_total":0
"modbus.selec.active_import_total":0
"modbus.selec.active_total":0
"modbus.selec.apparent_L1":0
"modbus.selec.apparent_L2":0
"modbus.selec.apparent_L3":0
"modbus.selec.apparent_total":0
"modbus.selec.current_L1":0
"modbus.selec.current_L2":0
"modbus.selec.current_L3":0
"modbus.selec.frequency":50
"modbus.selec.online":true
"modbus.selec.pf_L1":1
"modbus.selec.pf_L2":1
"modbus.selec.pf_L3":1
"modbus.selec.reactive_L1":0
"modbus.selec.reactive_L2":0
"modbus.selec.reactive_L3":0
"modbus.selec.reactive_export_total":0
"modbus.selec.reactive_import_total":0
"modbus.selec.reactive_total":0
"modbus.selec.voltage_L1":243.80999755859375
"modbus.selec.voltage_L1_L2":419.6300048828125
"modbus.selec.voltage_L1_L3":422.8999938964844
"modbus.selec.voltage_L2":241.97000122070312
"modbus.selec.voltage_L3":243.67999267578125
"modbus.selec.voltage_L3_L2":420.8999938964844
Total net KWH
Avg pF

- Check Dfun DC meter Communication
 - Check "modbus.dfun_dc.online":true
 - Check DC Parameter
 - CH1 - Energy , Power , voltage and Current Parameter
 - CH2 - Energy , Power , voltage and Current Parameter
 - Check all configure parameters shows in IO mapper

"modbus.dfun_dc.Alarm_bits_CH1" : 0

"modbus.dfun_dc.Alarm_bits_CH2" : 0

"modbus.dfun_dc.Current_CH1" : 0

"modbus.dfun_dc.Current_CH2" : 0

"modbus.dfun_dc.Current_forward_total_electric_energy_CH1" : 47.120002238079906

"modbus.dfun_dc.Current_forward_total_electric_energy_CH2" : 138.16000656224787

"modbus.dfun_dc.Current_reverse_total_electric_energy_CH1" : 0

"modbus.dfun_dc.Current_reverse_total_electric_energy_CH2" : 0

"modbus.dfun_dc.Power_CH1" : 0

"modbus.dfun_dc.Power_CH2" : 0

"modbus.dfun_dc.Voltage_CH1" : 0

"modbus.dfun_dc.Voltage_CH2" : 0

- Check Gongyuan IMD Communication
 - Check "modbus.gongyuan_IMD.online":true
 - Check IMD Parameter
 - Negative Pole-to-Ground Insulation Resistance: 60000 during cable check/charging
 1. 60000 means the measured resistance $\geq 10\text{M}\Omega$
 2. 65535 means the measured resistance $\leq 10\text{M}\Omega$

"modbus.gongyuan.Bus Voltage" : 0

"modbus.gongyuan.Negative Pole-to-Ground Insulation Resistance" : 65535

"modbus.gongyuan.Negative Pole-to-Ground Voltage" : 6553.5

"modbus.gongyuan.Positive Pole-to-Ground Insulation Resistance" : 65535

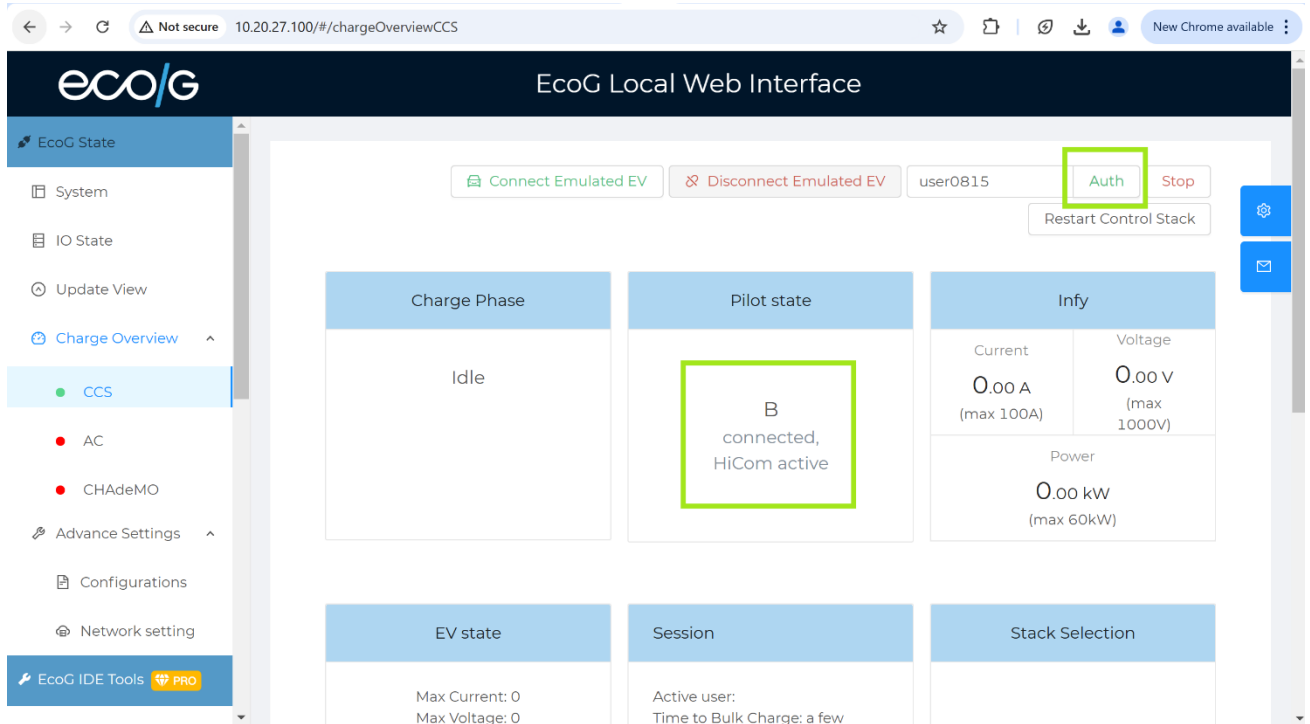
"modbus.gongyuan.Positive Pole-to-Ground Voltage" : 6553.5

"modbus.gongyuan.Status Bits" : 1045

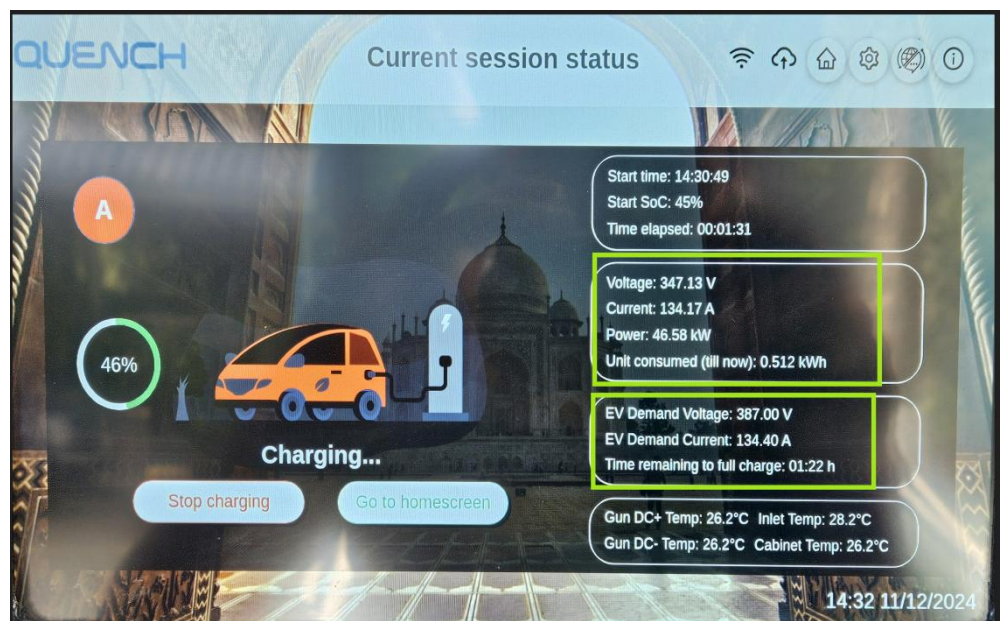
"modbus.gongyuan.Version Number" : 4098

4.4. Charging Test

- Verify AC DC Contactor and their control relay as per **Maintenance Mode Manual**
- Open LWI SECC “10.20.27.100”
 - Connect the Gun A to EV then check gun Pilot state (B connected, HiCom active)
 - Click on Auth to start the charging.



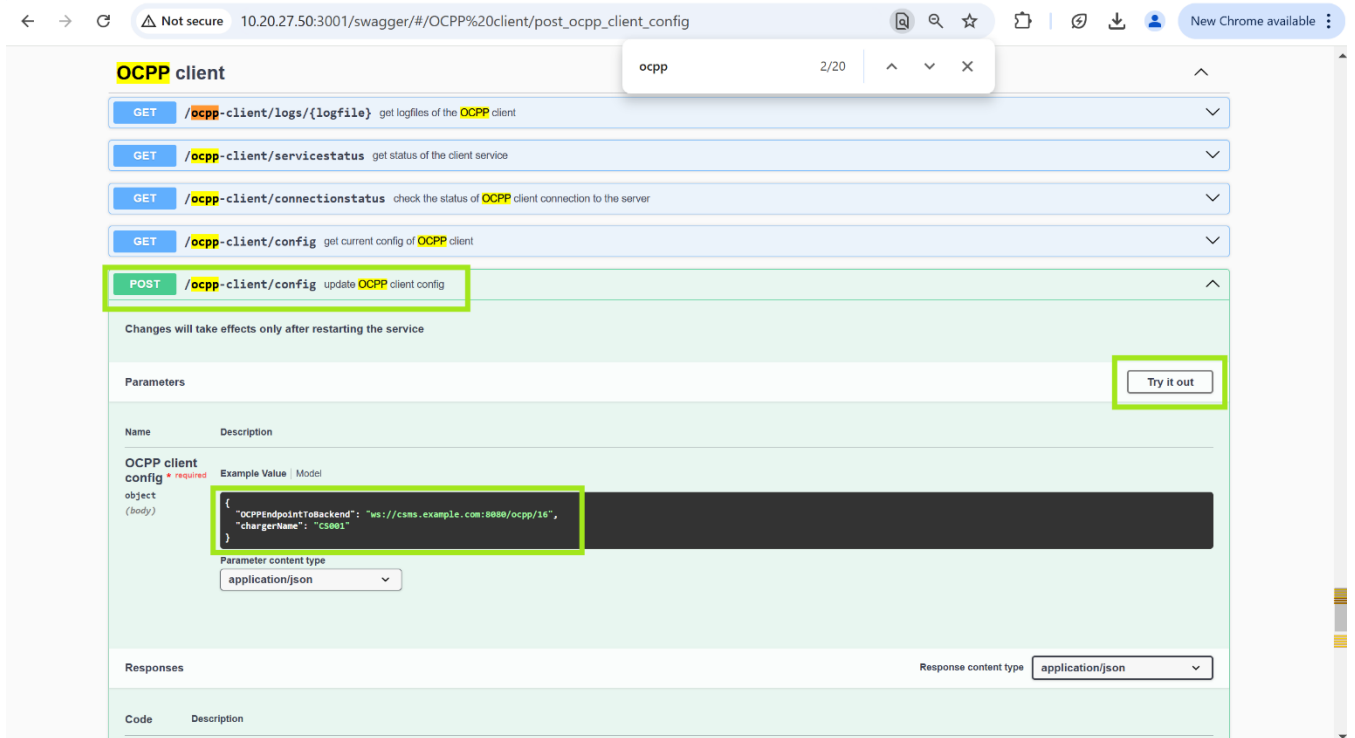
- Check EV and charger Parameter
 - 1) Check Battery SOC Display as per Vehicle Display
 - 2) Check EV demand current and supply current should be the same
 - 3) Check Display Functionality
 - 4) Check Charging start Process Cable check and pre-Charging
 - 5) Check Charging Stop Process Finishing



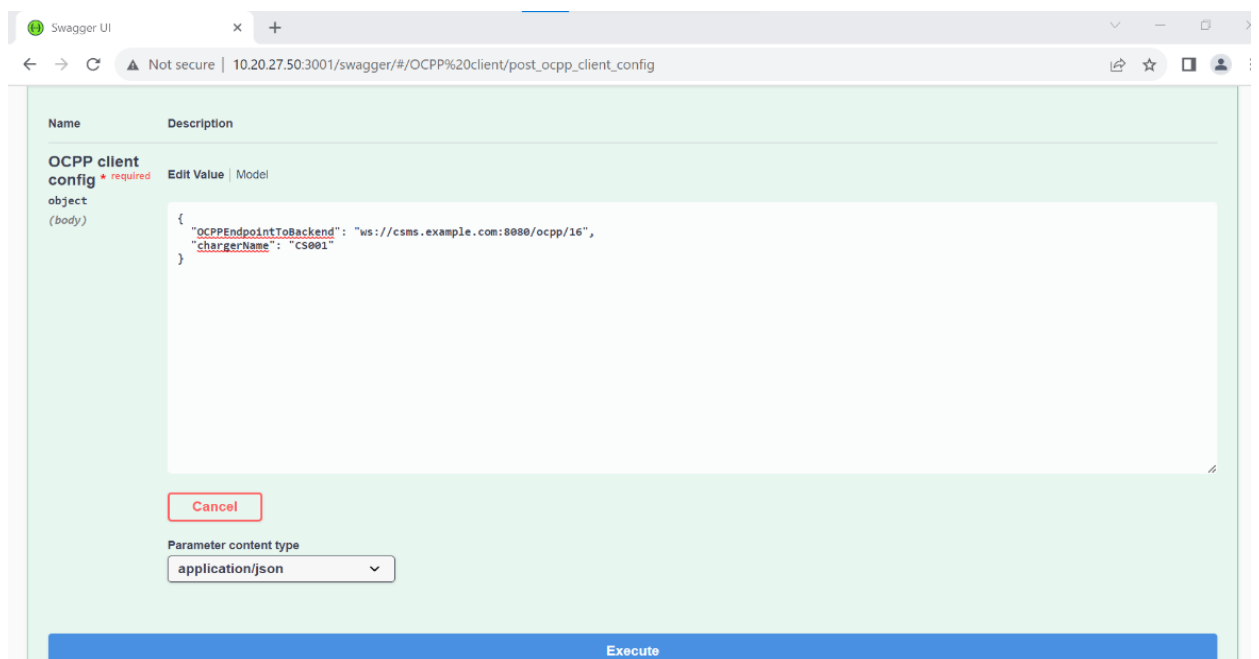
- Open LWI SECC-LE "10.20.27.101"
 - Perform same charging session on Gun B as mentioned above

4.5. OCPP Testing

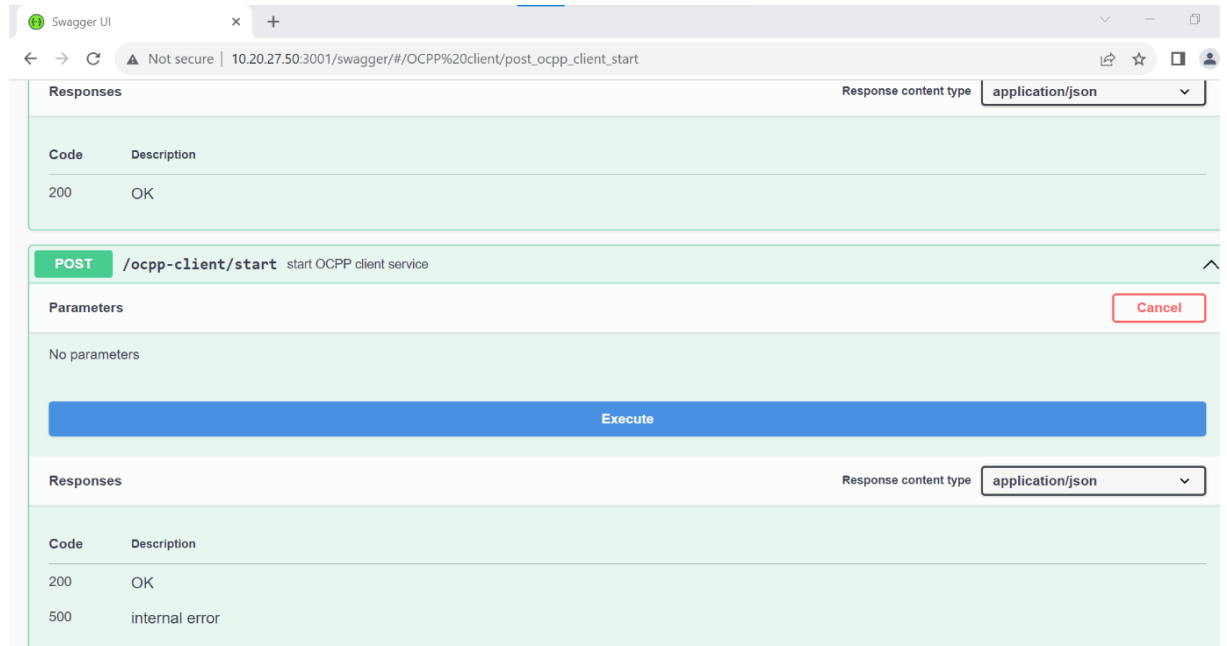
- Connect CMS
 - Open Swagger URL "10.20.27.50:3001"
 - Go to Post ocpp-client/Config



- Click on "POST ocpp-client/config". Click on Try it out. Paste the respective WebSocket URL and Charger ID and click on Execute.



- Click on "POST ocpp-client/start". Try it out and execute.



Swagger UI interface for the POST /ocpp-client/start endpoint. The interface shows the endpoint name, a description "start OCPP client service", and a table of responses with status codes 200 (OK) and 500 (internal error). A large blue "Execute" button is visible.

Code	Description
200	OK
500	internal error

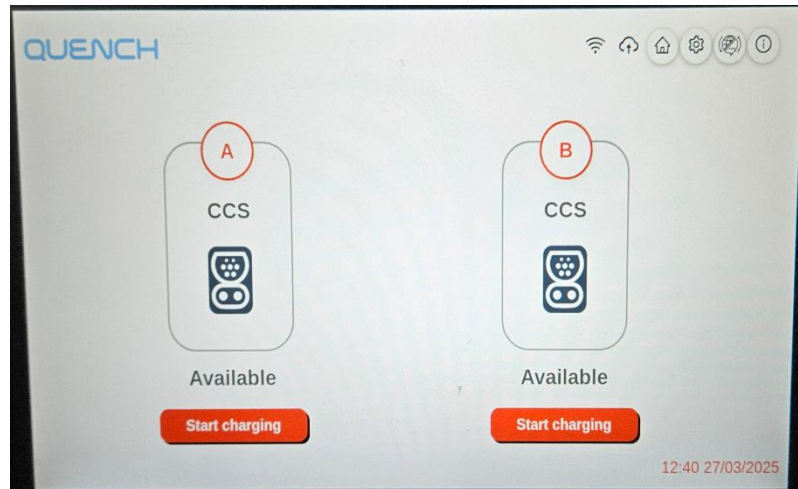
- Click on "POST ocpp-client/restart". Try it out and Execute.



Swagger UI interface for the POST /ocpp-client/restart endpoint. The interface shows the endpoint name, a description "restart OCPP client service", and a table of responses with status codes 200 (OK) and 500 (internal error). A large blue "Execute" button is visible.

Code	Description
200	OK
500	internal error

- The charger should now be connected to the CMS and the cloud sign will appear on the HMI.



- Open respective CMS URL in browser
 - There are 3 ways to authorize the charger
 - OTP
 - RFID
 - Remote Authorization
 - Test error codes applicable to idle state according to Error Codes List.
 - Check CMS Logs
 - Check screen behaviour
 - Reporting on HMI Alarms Logs (i button)
- Manual/Maintenance Mode allows the testing engineer to manually operate the DC contactor without PLC auto-control. This mode helps verify wiring correctness and ensures the DC contactor is switching properly during testing. Perform test as per **Maintenance Mode Manual**
- Perform charging session as per **OCPP testing document**
- Configure Offline OTP and 2 RFID in the Charger:
 - Connect Gun A to the EV or simulator.
 - Tap 2 RFID cards on the Charger RFID Reader to obtain the RFID numbers from the CMS log.
 - Open the Swagger URL: 10.20.27.50:3001
 - Connect to offline CMS:
 - Click on "POST ocpp-client/config". Then click on "Try it out" and execute.
 - Click on "POST ocpp-client/start". Then click on "Try it out" and execute.
 - Click on "POST ocpp-client/restart". Then click on "Try it out" and execute.
 - Click on "POST ocpp-client/config" again and click on "Try it out."
 - Edit the following code by entering the OTP and the two RFID numbers, then copy the code

```
"listVersion": 2,

"localAuthorizationList": [

{

  "idTag": "4732",

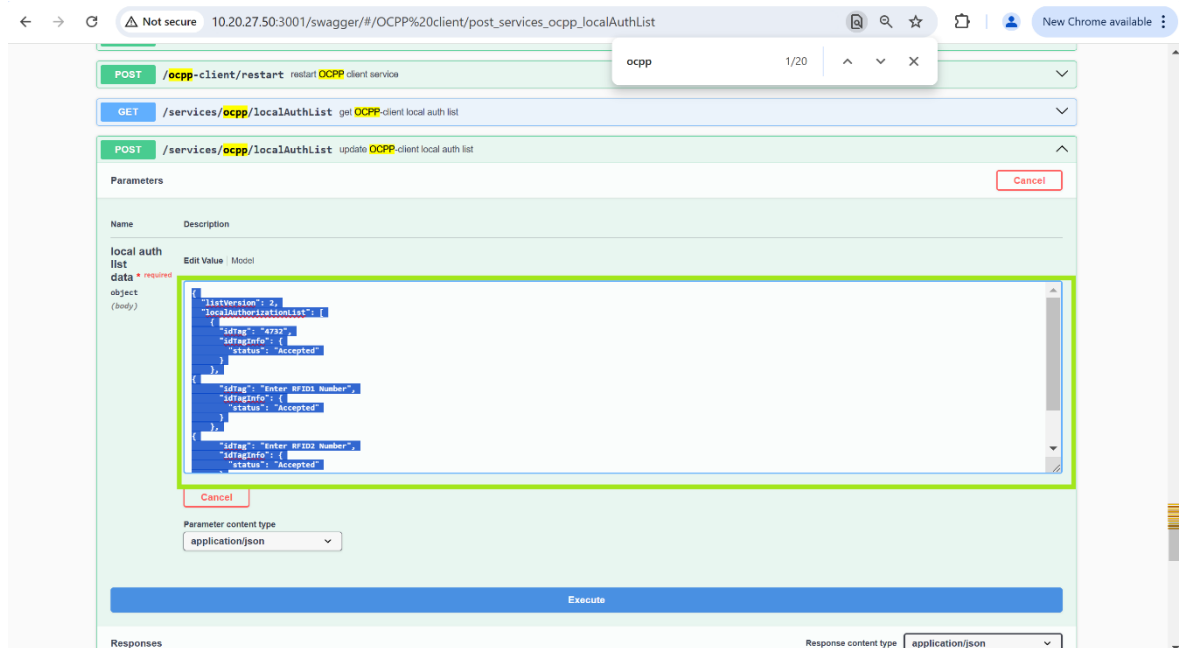
  "idTagInfo": {
```

```

        "status": "Accepted"
    }
},
{
    "idTag": "Enter RFID1 Number",
    "idTagInfo": {
        "status": "Accepted"
    }
},
{
    "idTag": "Enter RFID2 Number",
    "idTagInfo": {
        "status": "Accepted"
    }
}
],
"updateType": "Full"
}

```

- Paste the code into the highlighted area and execute.



- Click on "Get ocpp-client/config" and click on "Try it out."
- Check the uploaded code.

- After completing the OCPP test, reset the AC meter energy as per the AC meter manual above mention in page No. 13.
- After completing the OCPP test, reset the DC Meter energy as per below mentioned steps

Reset DC Meter:

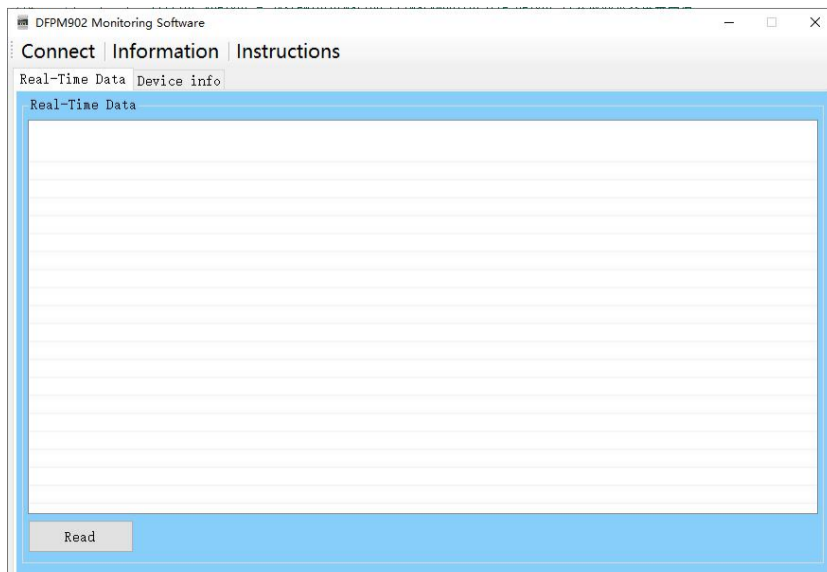
1. Connect your laptop to the DC Meter using RS485 to USB Converter.
2. To reset the DC Meter, download and install the debugging software sent by DC meter manufacturer, Dfun. The software is available on the server.
3. Once the software is installed, please open it and login using the following credentials:

User: admin

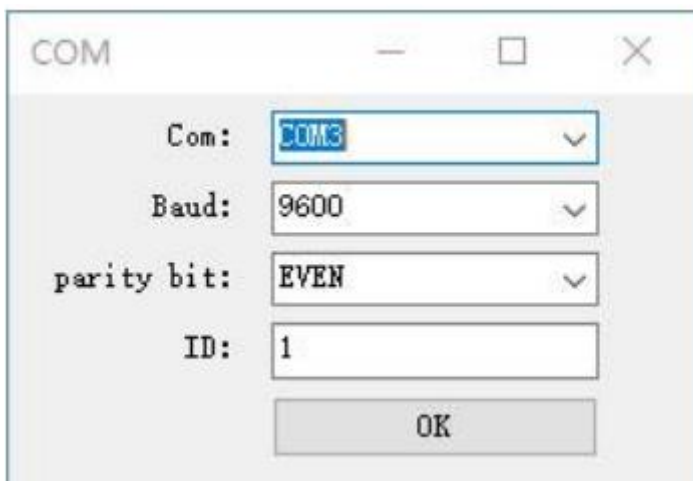
Password: admin



4. After logging in, the main interface will appear:



5. Click on the 'connect' button to configure the communication parameters. The following section will appear. Enter the same parameters as shown in the figure below. The parameter 'Com' might differ hence check the communication address mentioned in the device manager settings in the laptop.



6. After configuring the communication parameters correctly, click on the 'Read' button shown in the below to get the real-time data from the DC Meter.

-
- DFPM902 Monitoring Software
- Connect | Information | Instructions
- Real-Time Data: **Device info**
- Device Info:
- Device Time:
- Main Ver:
- Sub Ver:
- Beta:
- Boot Ver:
- SN:
- 645 Address:
- Update Read
- PPM calibration
- PPM:
- Reset Write
- Meter Clear
- Zero Clear
- Specification setting
- Loop1:
- Loop2:
- Read
- Set specifications---
- Loop:
- standards:
- Set
- Electric pulse output channel
- CH1 CH2
- Meter Type
- Read
- double loop Set
- Number of decimal places in electric energy
- Read
- Set
- Protocol
- Read
- standard Set

5.1. Authentication

- RFID card
- OTP
- Remote start

- Autocharge

The charging cycle will not initiate till the authentication is approved.

5.2. Pre-Charging Safety Checks

After authentication, the charger will run the safety check by checking any internal trips/alarms are present or not. Then it checks the external safety conditions like Charging gun faults, vehicle fault or any insulation fault present or any vehicle communication error.

If all safety condition passed, then only the actual charging can be initiated.

5.3. Charging

Before charging current start, the EVSE communicate with Vehicle BMS for the charging voltage and charging current requirement. The BMS decides the charging voltage and current as per Battery condition and rating. The controller then turns on the AC contactors and conveys the requirement to the power modules via CAN communication. Subsequently, the controllers turn on the respective DC Contactors as the power modules ramp up to the required voltage for cable check. After successful cable check and pre-charge stages, the power module sets its output as per requirements shared by the vehicle and power delivery to the vehicle begins. The charging voltage/current including battery SOC are continuously monitored and displayed on screen so that user can get the charging status at any given time.

5.4. Charging completion:

The charger is continually sharing the charging status of vehicle and the respective charges to user via mobile application. The Charging process will stop if any of the following conditions occur:

- The battery is fully charged.
- Preset energy /time duration/Final SOC requirement
- The STOP button pressed to discontinue the charging.
- If emergency stop button pressed.
- If by any means, if charging process completed/stopped then Charger will initiate the billing process as per last power fed to vehicle. The billing is conveyed to user on mobile application and on HMI Display.
- The User need to initiate new charger cycle to start charging again.

5.5. Error/Fault, interrupt or program stop condition

During an error, fault, interrupt, or program stop (program finished) condition the power module's outputs are stopped and the AC and DC contactors are opened.

The billing cycle will be shown in completion state when system Trips or Charging finished or forcefully stopped.